Olti Myrtaj

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RESEARCH INTERESTS

Galaxy formation and the dynamical origins of galaxy disks: the Milky Way's stellar archaeology and the earliest rotating galaxies. The gravitational and feedback-driven effects of massive black holes on their host galaxies. Cosmological hydrodynamic simulations, galactic dynamics, and mock ALMA/JWST observations of galaxy kinematics/morphology. Cosmological tensions: early dark energy models and the Hubble tension. Small-scale structure of the Milky Way: self-interacting dark matter, too-big-to-fail, missing satellites.

UAT keywords: Galaxy formation (595) · Galaxy kinematics (602) · High-redshift galaxies (734)

EDUCATION

2021–27 **Ph.D., Physics (Astrophysics)**, University of California, Irvine; expected June 2027.
Advisor: James S. Bullock. Working thesis title: *The Three-phase Kinematic Evolution and Physical Mechanisms of Disk Structure Formation in Milky Way-mass Galaxies across Cosmic Time*.
Qualifying examination: authored and defended a 25-page literature review, *Early Dark Energy Models and their Ability to Alleviate the Hubble Tension*; passed with outstanding marks before a committee of three professors.

2024 **M.S., Physics**, University of California, Irvine (en route to the Ph.D.).

2021 **B.S., Physics**; minor in Mathematics, Virginia Tech.
Physics Valedictorian (4.0 GPA); H. Y. Loh Award; *summa cum laude*; Phi Beta Kappa; Dean's Roundtable Scholar.

PUBLICATIONS

First author

Myrtaj, O., Bullock, J. S., Boylan-Kolchin, M., Chandra, V., Faucher-Giguère, C.-A., Feldmann, R., Gandhi, P. J., Mercado, F. J., Moreno, J., Stern, J., Wetzel, A., & Xia, L. Y. 2026, "From protogalaxy through thick and thin: Why did the Milky Way evolve in three kinematic phases?", *MNRAS*, accepted; arXiv:2605.22806.

Myrtaj, O., et al., "Turning Point: The Dynamical Origin of the Milky Way's Initial Disk and Angular Momentum Structure." In preparation.

Myrtaj, O., et al., "Disks in Progress: The Incidence and Observability of Gas Disks Without Stellar Disks in FIREbox and MassiveFIRE." In preparation.

Contributing author

Bian, S., Feldmann, R., Faucher-Giguère, C.-A., Anglés-Alcázar, D., Moreno, J., & **Myrtaj, O.**, "MassiveFIRE-2: A characteristic mass for stellar disc formation at high redshift." In preparation.

Rinaldi, S., Wellons, S., Wetzel, A., Sultan, I., Steel, C., Anglés-Alcázar, D., Benavides, J. A., Moreno, J., **Myrtaj, O.**, & Xia, L. Y., "The Emergence of Massive Stellar Disks at High Redshift in FIRE-2 Simulations: Linking Morphology and Kinematics from Cosmic Dawn to Cosmic Noon." In preparation.

RESEARCH EXPERIENCE

Graduate Student Researcher, University of California, Irvine 2021–present

- **First-author research on galactic disk formation:** led a study of twelve FIRE-2 cosmological zoom-in simulations to explain how Milky Way-mass galaxies evolve from disordered protogalaxies into thick and thin disks. Quantified gas and stellar orbital circularities over cosmic time and connected their evolution to the three phases recorded in Gaia/APOGEE stellar archaeology (Myrtaj et al. 2026, *MNRAS*, **accepted**).
- **Original measure of baryonic settling:** introduced a new foundational framework for disk structure, applying a dimensionless measure of baryonic "sloshing," the bulk velocity offset of a galaxy's gas and stars relative to its total matter, normalized by circular velocity. Identified declining sloshing as the clearest dynamical change associated with the formation of coherent rotation, linking the formation of disk structure to the emergence of a stable dynamical center.
- **Distinct stages of disk formation:** separated the onset of coherent rotation from the later birth of stars on dynamically cold thin-disk orbits. Linked this later transition to steady star formation and inner circumgalactic-gas virialization; found that the rate of cool-gas conversion into stars, relative to the available cool-gas mass, peaks during the intervening thick-disk phase, while star formation remains bursty.
- **Ruling out competing physical explanations:** evaluated mass growth, mergers, gas fractions and the shape of the central gravitational potential against the timing of disk onset. Found no systematic merger trigger or single stellar-mass, halo-mass or gas-fraction threshold for spin-up across the sample. The central potential remains relatively unconcentrated when coherent gas rotation first emerges.
- **Observational predictions for gas and stellar kinematics:** established that cool gas begins to systematically rotate before stars, with delays ranging from a few hundred Myr to more than 3 Gyr. Identified an observable intermediate phase with an ordered gas velocity field and dynamically hot young stars, showing why gas rotation alone does not establish that a galaxy has formed a coherent, rotation-supported stellar disk.
- **Research software and methods:** designed and built the Python/HDF5 pipeline supporting the study, and maintain its parallel GIZMO snapshot reduction, orbital-analysis and publication-figure tools on UCI's HPC3 cluster (Slurm). Integrated temperature-resolved gas kinematics, stellar age and metallicity distributions, baryonic bulk motions, rotation curves and circumgalactic cooling times to test the physical interpretation across the full simulation suite.

RESEARCH IN PROGRESS

- **The dynamical mechanism of disk spin-up:** leading a first-author investigation of gravitational torques and the evolving galactic potential, using potential multipoles and tidal tensors. Manuscript: *Turning Point: The Dynamical Origin of the Milky Way's Initial Disk and Angular Momentum Structure* (in preparation).
- **Gas disks before stellar disks:** leading a first-author study of how common galaxies with a gas disk but no stellar disk are across redshift and stellar mass, from FIREbox's cosmological volume to MassiveFIRE's high-mass galaxies. Reduced the data, built mock ALMA [C II] and JWST NIRCam pipelines and identified prototype galaxies, to determine from their gas and stellar orbits which stage of disk formation each telescope captures. Manuscript: *Disks in Progress* (in preparation).
- **Gravity, stellar feedback and early disk formation:** independently tested a disk-formation criterion comparing gravitational binding with stellar-feedback energy on twelve Milky Way-mass FIRE-2 galaxies, including alternative gas selections. This contributes to a MassiveFIRE-2 study of the characteristic stellar mass for disk formation at $z = 4$–5 (Bian, Feldmann et al.; manuscript in preparation).

- **The emergence of massive stellar disks:** contributed to the physical interpretation of a MassiveFIRE study tracking galaxy shapes and stellar rotation from $z = 7$ to $z = 1$; connected the results to my work on gas sloshing and thick- and thin-disk formation, and proposed tests of central mass concentration and baryonic motion (Rinaldi, Wellons et al.; manuscript in preparation).

COMPUTING & SCIENTIFIC METHODS

- **Simulation analysis:** FIRE-2 cosmological zoom-ins (Latte, ELVIS), FIREbox and MassiveFIRE; GIZMO snapshots and Rockstar halo catalogs; parallel reduction and HDF5 data products on HPC clusters; mock ALMA [C II] and JWST NIRCam observations.
- **Dynamics & gas physics:** orbital circularity, angular momentum and gas/stellar kinematics; circumgalactic gas thermodynamics; potential multipoles, tidal tensors and gravitational torques.
- **Software & statistics:** Python (NumPy, SciPy, h5py, Astropy, scikit-learn), Mathematica, Bash, Git; random forests and Gaussian mixture models.
- **Observational training:** Armin O. Leuschner Workshop, Lick Observatory (2024); optical spectroscopy (Shane/Kast, CAT/Hamilton) and imaging (Nickel 1-m).

SELECTED TALKS & PRESENTATIONS

Jan 2027	"The Physical Origins of the Milky Way's Disk" 249th AAS Meeting, Salt Lake City, Utah (contributed talk).
Oct 2026	"The Physical Drivers of the Milky Way's Disk Formation." CosmoLab Seminar Series, University of Southern California (invited talk).
Sep 2026	"The Physical Drivers of the Milky Way's Disk Formation." Galaxy Formation and Evolution in Southern California (GalFRESCA), University of California, Irvine (talk).
Jun 2025	"Protogalaxy, thick disk, thin disk: Why did the Milky Way evolve in three kinematic phases?" 246th AAS Meeting, Anchorage, Alaska (contributed talk).
Sep 2024	"Why did the Milky Way spin up?"GalFRESCA, Carnegie Observatories, Pasadena (contributed talk).
Jul 2024	"The Three-phase Kinematic Structure of the Milky Way." Santa Cruz Galaxy Workshop, UC Santa Cruz (talk).
Jul 2024	"Why did the Milky Way Spin Up?" European Astronomical Society Annual Meeting, Padova, Italy (poster).
Feb 2024	"What causes the formation of disks?" University of California, Los Angeles (invited talk).
Sep 2023	"The Three-phase Structure of the Milky Way's Kinematic History." GalFRESCA, University of California, Riverside (contributed talk).
Sep 2023	"Un-concentrated: the galaxy potential and its lacking role in forming disks." GalFRESCA, California Institute of Technology (contributed talk).
Jun 2022	"The Most Massive Subhaloes of Milky Way-mass Galaxies in Self-Interacting Dark Matter Models." 240th AAS Meeting, Pasadena (contributed talk).

SCIENTIFIC ORGANIZATION & SERVICE

2026	**Co-organizer and local host, GalFRESCA**, UC Irvine: two-day regional galaxy-formation meeting; secured meeting funding; co-organized with T. Smith (USC) and L. Sandoval Ascencio (UCI).
2024–26	**Scientific Organizing Committee, Sierra Conference**, UC-wide graduate astronomy meeting; UC Irvine representative.
2024	**Local organizer, GalRead**, galaxy-formation reading group, UCLA.

2021– **Research-group coordination**: moderate weekly meetings; manage the shared FIRE simulation archive and analysis-code repository.

RESEARCH SUPERVISION & MENTORING

2025–26 **Undergraduate research supervision**, UCI CAMP summer research program: supervised Evelyn Castaneda's project, presented as a poster at the 247th AAS Meeting (January 2026).

2023– **Graduate research training**: trained two incoming graduate students in the group's FIRE analysis pipeline and HPC workflows.

2023–26 **Graduate Student Leader, SURF**, UC Irvine: lead graduate mentor for 32 visiting undergraduate researchers over four summers (eight per summer), supporting research progress and professional development; co-organized the annual symposium with 80–90 presenters.

2025–27 **Graduate Mentor, UC LEADS**, UC Irvine: mentored 16 scholars in 2025–26 through research communication and graduate-school and fellowship applications; reappointed for 2026–27.

2025–26 **Peer Mentor, Competitive Edge**, UC Irvine: supported eight incoming Ph.D. students over two summers.

2026 **Research Facilitator, COSMOS**, UC Irvine: guided first physics research projects for 30 high-school students.

TEACHING

2021–23 **Teaching Assistant**, UC Irvine: six courses over six quarters, including introductory physics, laboratory instruction and Modern Physics for Majors.

2023–26 **Workshops led**: *responsible AI use in academic research* (UC LEADS), research abstracts, posters and statements of purpose (SURF); graduate readiness, scientific writing.

2018–21 **Teaching Assistant and Grader**, Virginia Tech: Mathematical Methods of Physics; Differential Equations.

SCIENCE COMMUNICATION & OUTREACH

2024–25 **Science Communication Fellow**, UCI School of Physical Sciences (one of four graduate fellows selected): wrote UCI News features and a press release for an NSF Major Research Instrumentation award.

2024–25 **Project lead, Discovery Cube Orange County Solar System exhibit**: led eight graduate students in scientific content and design review.

2026 **Facilitator, Rising Star Girls**: astronomy-and-art program for 50 participants.

2025 **Staff astronomer, Joshua Tree Dark Sky Festival**: public observing and astronomy communication (approximately 100 participants).

EARLIER RESEARCH

2020–21 **NSF REU, Astronomy**, UT Austin; O. Sameie and M. Boylan-Kolchin. Compared Milky Way satellite abundances, circular-velocity profiles and central densities in FIRE simulations of cold and self-interacting dark matter, with and without baryonic physics.

2019–21 **Undergraduate research**, Virginia Tech; I. Shoemaker. Investigated hidden-charged dark matter and dark matter–neutrino interactions, with implications for small-scale galactic structure, using Mathematica.

2019 **NSF REU, Mathematics**, Indiana University; C. Judge. With K. Chen, proved the existence of a degenerate, index-zero critical point of a Dirichlet Laplacian eigenfunction on the square; numerically located it and visualized its behavior in Mathematica.